

AI Medicine Price Predictor

Smart AI system for predicting medicine price category





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Problem Statements :

- 1.** Lack of Price Awareness
- 2.** Variation in Medicine Pricing
- 3.** Time-Consuming Comparison
- 4.** Lack of Smart Prediction Systems
- 5.** Need for Automation
- 6.** Accessibility Issue

Real-World Challenges :

- 1.** Price Variation in Medicines
- 2.** Lack of Price Transparency
- 3.** Complex Decision-Making
- 4.** Limited Awareness
- 5.** Time-Consuming Manual Analysis
- 6.** No Smart Prediction Systems
- 7.** Risk of Overpaying





Objectives :

- To develop a Machine Learning model to predict the price category of medicines
- To analyze how features like manufacturer, type, and composition affect pricing
- To build a user-friendly web application using Streamlit
- To provide quick and data-driven predictions for users
- To demonstrate the practical use of Machine Learning in real-world problems



Technologies Used :

- Python
- Machine Learning
- Streamlit
- Pandas, NumPy, Scikit-learn



Dataset Explanation :

The dataset used in this project contains information about different medicines along with their key features such as manufacturer, type, and composition. Each record is labeled with a corresponding price category (e.g., High or Low), which is used as the target variable for prediction.

The dataset was preprocessed and cleaned to ensure accuracy and consistency before training the machine learning model.



Data Preprocessing :

- Cleaned dataset
- Removed missing or inconsistent values
- Encoded categorical features into numerical format
- Structured data for machine learning model
- Split into training and testing data



Machine Learning Model :

In this project, a Machine Learning classification model is used to predict the price category of medicines based on input features such as manufacturer, type, and composition. Multiple machine learning models were used to predict the price category of medicines. This helps in comparing performance and selecting the most accurate model.

- **Used 3 models :**

- Random Forest

- Gradient Boosting

- Logistic Regression

- Compared performance
- Selected best model



System Workflow :

The system follows a structured pipeline where user input is processed and passed through a trained machine learning model to generate the final prediction.

Input → Preprocessing → Model → Prediction → Output

Application :

- Built using Streamlit
- User inputs features
- Model predicts price category
- Displays instant result

Medicine AI

Enter Details

Manufacturer

Sun Pharmaceutical Industries Ltd

Type

Tablet

Choose Composition

Select from list

Composition

Amoxycillin (500mg) Clavulanic ...

Predict

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Model Type	Features Used	Output
ML Pipeline	3	Price Category

Built with Streamlit | AI Project

Prediction Output :

- Predicts High / Low price category
- Instant result display
- Easy to understand output

Medicine AI

Enter Details

Manufacturer

Dr Reddy's Laboratories Ltd

Type

Tablet

Choose Composition

Select from list

Composition

Hydroxyzine (10mg) nan

Predict

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Prediction Result

HIGH Price Range

Expensive medicine

Model Type	Features Used	Output
ML Pipeline	3	Price Category



Features :

- Real-time prediction
- Easy-to-use interface
- Fast performance
- Multiple ML models
- Feature-based prediction



Advantages :

- Fast and efficient
- Easy to use
- Saves time
- Data-driven predictions
- Accessible to everyone

Limitations :

- Dataset dependent
- Limited features
- Predicts category only
- Not 100% accurate
- No real-time data



Future Scope :

- Add more features
- Improve accuracy
- Predict exact price
- Use real-time data
- Deploy online



Conclusion:

- This project successfully demonstrates the application of Machine Learning in predicting the price category of medicines based on key features such as manufacturer, type, and composition.
- By integrating the model into a user-friendly web application using Streamlit, the system provides quick and accessible predictions.
- Overall, the project highlights how data-driven approaches can simplify complex decision-making in real-world scenarios.

“This project shows the potential of machine learning in making intelligent and accessible prediction systems.”

